

nD: 4, H: 1, H+: 1

term0

$$-\left(\frac{1}{24}a_t^2 - \frac{1}{4}a_t\hat{\mu} + \frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} - \frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon^{\alpha_4\alpha_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_3}D_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}H_iD_{\alpha_3}^{\dot{\alpha}_3}D_{\alpha_4}^{\dot{\alpha}_4}(H^\dagger)_j \quad (1)$$

term3

$$+\left(\frac{1}{4}a_t\hat{\mu} - \frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_3\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_4}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}D_{\alpha_3}^{\dot{\alpha}_3}D_{\alpha_4}^{\dot{\alpha}_4}(H^\dagger)_j \quad (2)$$

term4

$$+\left(\frac{1}{4}a_t\hat{\mu} - \frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_3\alpha_2}\varepsilon^{\alpha_1\alpha_4}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_4}D_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}D_{\alpha_3}^{\dot{\alpha}_3}H_iD_{\alpha_4}^{\dot{\alpha}_4}(H^\dagger)_j \quad (3)$$

term5

$$+\left(\frac{1}{2}a_t\hat{\mu} - \frac{1}{2}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_4}\varepsilon^{\alpha_1\alpha_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_4}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_3}D_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}H_iD_{\alpha_3}^{\dot{\alpha}_3}D_{\alpha_4}^{\dot{\alpha}_4}(H^\dagger)_j \quad (4)$$

term6

$$+\left(\frac{1}{8}a_t\hat{\mu} - \frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{1}{4}\frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_4\alpha_3}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}D_{\alpha_3}^{\dot{\alpha}_3}D_{\alpha_4}^{\dot{\alpha}_4}H_i(H^\dagger)_j \quad (5)$$

term7

$$+\left(\frac{1}{4}a_t\hat{\mu} - \frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_3\alpha_4}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_4}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}D_{\alpha_3}^{\dot{\alpha}_3}H_iD_{\alpha_4}^{\dot{\alpha}_4}(H^\dagger)_j \quad (6)$$

term8

$$+\left(\frac{1}{4}a_t\hat{\mu} - \frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon^{\alpha_4\alpha_3}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_3}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}D_{\alpha_3}^{\dot{\alpha}_3}D_{\alpha_4}^{\dot{\alpha}_4}(H^\dagger)_j \quad (7)$$

term10

$$+\left(\frac{1}{8}a_t\hat{\mu} - \frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{1}{4}\frac{a_t\hat{\mu}}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_4\alpha_3}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iD_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}D_{\alpha_3}^{\dot{\alpha}_3}D_{\alpha_4}^{\dot{\alpha}_4}(H^\dagger)_j \quad (8)$$

nD: 2, H: 1, H+: 1

term1

$$+\left(\frac{1}{4}a_t^2 + \frac{1}{2}a_t\hat{\mu} - \frac{1}{2}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} - \frac{1}{2}\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 + \frac{5}{4}a_t^2\hat{\mu}_h^2 + \frac{a_t\hat{\mu}}{\varepsilon} + \frac{a_t^2\hat{\mu}_h^2}{\varepsilon}\right)\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j \quad (9)$$

term2

$$+\left(\frac{1}{4}a_t\hat{\mu} - \frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu} + \frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right)\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}H_i(H^\dagger)_j \quad (10)$$

term9

$$+\left(\frac{1}{4}a_t\hat{\mu}-\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu}+\frac{1}{2}\frac{a_t\hat{\mu}}{\varepsilon}\right)\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iD_{\alpha_1}^{\dot{\alpha}_1}D_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j \quad (11)$$

nD: 0, H: 1, H+: 1

term11

$$\begin{aligned} & -\left(a_t^2+a_t^2\hat{\mu}_h^2+a_t^2\hat{\mu}_h^4-\ln\frac{M_s^2}{\mu_M^2}a_t^2-\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2-\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^4-\frac{1}{2}a_t\hat{\mu}+\frac{1}{2}\ln\frac{M_s^2}{\mu_M^2}a_t\hat{\mu}+\frac{1}{2}\kappa\right. \\ & \left.-\frac{1}{2}\kappa\ln\frac{M_s^2}{\mu_M^2}+2\frac{a_t^2}{\varepsilon}+2\frac{a_t^2\hat{\mu}_h^2}{\varepsilon}+2\frac{a_t^2\hat{\mu}_h^4}{\varepsilon}-\frac{a_t\hat{\mu}}{\varepsilon}+\frac{\kappa}{\varepsilon}\right)M_s^2\varepsilon_{ij}H_i(H^\dagger)_j \end{aligned} \quad (12)$$

nD: 2, BL: 1, H: 1, H+: 1

term12

$$+\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (13)$$

term12

$$+\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (14)$$

term13

$$+\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (15)$$

term13

$$+\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(B_L)_{\alpha_1\alpha_2}D_{\alpha_3}^{\dot{\alpha}_1}H_iD_{\alpha_4}^{\dot{\alpha}_2}(H^\dagger)_j \quad (16)$$

nD: 2, H: 1, H+: 1, BL+: 1

term12

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_1}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j(B_R)^{\dot{\alpha}_3\dot{\alpha}_4} \quad (17)$$

term12

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_2}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j(B_R)^{\dot{\alpha}_3\dot{\alpha}_4} \quad (18)$$

term13

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_1}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j(B_R)^{\dot{\alpha}_3\dot{\alpha}_4} \quad (19)$$

term13

$$-\left(\frac{1}{96}g_1\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_1a_t^2}{\varepsilon}-\frac{17}{576}g_1a_t^2\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_3\dot{\alpha}_2}\varepsilon_{\dot{\alpha}_4\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}H_iD_{\alpha_2}^{\dot{\alpha}_2}(H^\dagger)_j(B_R)^{\dot{\alpha}_3\dot{\alpha}_4} \quad (20)$$

nD: 2, H: 2, H+: 2

term14

$$\begin{aligned}
& + \left(a_t^3 \hat{\mu} \hat{\mu}_h^2 - a_t^4 \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 - \kappa a_t^2 - \kappa a_t^2 \hat{\mu}_h^2 - \kappa a_t^2 \hat{\mu}_h^4 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 \right. \\
& + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 + \frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 + \frac{1}{8} \lambda_\varphi a_t^2 - \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 \\
& - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} - \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} + \frac{1}{2} a_t^3 \hat{\mu} - \frac{1}{2} a_t^4 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 \\
& \left. + \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{2} \kappa a_t \hat{\mu} + \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} - 2 \frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H_i H_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{21}$$

term15

$$\begin{aligned}
& - \left(\frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{8} \lambda_\varphi a_t^2 + \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{4} a_t^2 \hat{\mu}^2 \right. \\
& \left. - \frac{1}{4} \kappa a_t \hat{\mu} + \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} - \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} - \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H_i H_j \\
& (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{22}$$

term16

$$\begin{aligned}
& + \left(a_t^3 \hat{\mu} \hat{\mu}_h^2 - a_t^4 \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 - \kappa a_t^2 - \kappa a_t^2 \hat{\mu}_h^2 - \kappa a_t^2 \hat{\mu}_h^4 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 \right. \\
& + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 + \frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 + \frac{1}{8} \lambda_\varphi a_t^2 - \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 \\
& - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} - \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} + \frac{1}{2} a_t^3 \hat{\mu} - \frac{1}{2} a_t^4 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 \\
& \left. + \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{2} \kappa a_t \hat{\mu} + \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} - 2 \frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \right) M_s^{-2} \\
& \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{23}$$

term17

$$\begin{aligned}
& - \left(\frac{3}{32} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{8} \lambda_\varphi a_t^2 + \frac{1}{8} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{9}{64} g_2^2 a_t^2 - \frac{3}{16} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{4} a_t^2 \hat{\mu}^2 \right. \\
& \left. - \frac{1}{4} \kappa a_t \hat{\mu} + \frac{1}{4} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \frac{1}{4} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} - \frac{1}{4} \frac{\lambda_\varphi a_t^2}{\varepsilon} - \frac{1}{2} \frac{\kappa a_t \hat{\mu}}{\varepsilon} \right) M_s^{-2} \varepsilon_{jk} \varepsilon_{il} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j \\
& D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{24}$$

term18

$$\begin{aligned}
& - \left(a_t^2 \hat{\mu}^2 + \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + \frac{1}{24} d g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{144} d g_2^2 a_t^2 - \frac{1}{12} \frac{d g_2^2 a_t^2}{\varepsilon} \right. \\
& \left. - \frac{1}{8} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} + \frac{1}{8} \varepsilon \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} \varepsilon a_t^3 \hat{\mu} - \frac{1}{4} \varepsilon \kappa a_t^2 - \frac{29}{72} g_2^2 a_t^2 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 \right)
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2}\kappa a_t^2 \hat{\mu}_h^2 - \frac{1}{2}\lambda_\varphi a_t^2 + \frac{1}{2}\lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{2}g_2^2 a_t^2 \hat{\mu}_h^2 + \frac{1}{2}g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{2}\lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 \\
& + \frac{1}{2}\varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{7}{12}g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{4}a_t^3 \hat{\mu} - \frac{3}{4}\varepsilon a_t^4 - \frac{7}{6}\frac{g_2^2 a_t^2}{\varepsilon} - \frac{5}{4}a_t^4 + \frac{5}{4}\kappa a_t^2 - \frac{3}{2}\kappa a_t \hat{\mu} + \frac{9}{4}\lambda_h a_t^2 \\
& - 2a_t^3 \hat{\mu} \hat{\mu}_h^2 + 2a_t^4 \hat{\mu}_h^2 - 2\ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 + 2\kappa a_t^2 \hat{\mu}_h^4 - 2\kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 - 2\kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 - 2\frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} \\
& - 2\frac{\kappa a_t \hat{\mu}}{\varepsilon} + 2\frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + 4\frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} + 4\frac{\kappa a_t^2}{\varepsilon} + 4\frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{a_t^4}{\varepsilon} - \frac{\lambda_\varphi a_t^2}{\varepsilon} - \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \\
& + \frac{\lambda_h a_t^2}{\varepsilon} \Big) M_s^{-2} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} H_i H_j D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{25}$$

term19

$$\begin{aligned}
& + \left(\frac{1}{96} dg_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_1^2 a_t^2 - \frac{1}{48} \kappa^2 + \frac{1}{48} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_1^2 a_t^2}{\varepsilon} - \frac{11}{288} dg_2^2 a_t^2 - \frac{1}{24} g_1^2 a_t^2 \right. \\
& - \frac{1}{24} \frac{dg_2^2 a_t^2}{\varepsilon} - \frac{5}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{5}{48} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{8} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{8} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{8} g_1^2 a_t^2 \hat{\mu}_h^2 \\
& + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{6} a_t^3 \hat{\mu} - \frac{3}{16} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{121}{576} g_2^2 a_t^2 + \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 \\
& - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{4} \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - \frac{13}{48} a_t^2 \hat{\mu}^2 + \frac{11}{32} \varepsilon a_t^4 + \frac{5}{12} \kappa a_t^2 \\
& - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{2} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{1}{2} \frac{a_t^4}{\varepsilon} - \frac{1}{2} \frac{\lambda_h a_t^2}{\varepsilon} + \frac{13}{24} \kappa a_t \hat{\mu} - \frac{7}{8} \lambda_h a_t^2 + \frac{7}{6} a_t^4 \\
& \left. + \frac{5}{4} \kappa a_t^2 \hat{\mu}_h^2 + \frac{\kappa a_t \hat{\mu}}{\varepsilon} + \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} H_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{26}$$

term20

$$\begin{aligned}
& - \left(\kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} + \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{48} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{48} dg_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{11}{288} dg_2^2 a_t^2 - \frac{11}{288} dg_1^2 a_t^2 \right. \\
& + \frac{1}{24} \kappa^2 + \frac{1}{24} \frac{dg_2^2 a_t^2}{\varepsilon} - \frac{1}{24} \frac{dg_1^2 a_t^2}{\varepsilon} + \frac{29}{144} g_2^2 a_t^2 - \frac{29}{144} g_1^2 a_t^2 + \frac{1}{4} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{4} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 \\
& - \frac{1}{4} g_1^2 a_t^2 \hat{\mu}_h^2 + \frac{1}{4} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{7}{24} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{7}{24} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{3} a_t^3 \hat{\mu} + \frac{3}{8} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 \\
& - \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 + \frac{1}{2} \ln \frac{M_s^2}{\mu_M^2} a_t^4 - \frac{1}{2} \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{2} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - \frac{1}{2} \frac{g_1^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} + \frac{13}{24} a_t^2 \hat{\mu}^2 + \frac{7}{12} \frac{g_2^2 a_t^2}{\varepsilon} \\
& - \frac{7}{12} \frac{g_1^2 a_t^2}{\varepsilon} - \frac{11}{16} \varepsilon a_t^4 - \frac{5}{6} \kappa a_t^2 - \frac{13}{12} \kappa a_t \hat{\mu} + \frac{9}{4} \lambda_h a_t^2 - \frac{7}{3} a_t^4 - \frac{5}{2} \kappa a_t^2 \hat{\mu}_h^2 - 2 \frac{\kappa a_t \hat{\mu}}{\varepsilon} - 2 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + \frac{a_t^2 \hat{\mu}^2}{\varepsilon} \\
& \left. - \frac{a_t^4}{\varepsilon} + \frac{\lambda_h a_t^2}{\varepsilon} \right) M_s^{-2} \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} H_i H_j D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{27}$$

term21

$$\begin{aligned}
& - \left(\frac{1}{96} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_2^2 a_t^2 - \frac{1}{48} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_2^2 a_t^2}{\varepsilon} + \frac{1}{24} \frac{g_1^2 a_t^2}{\varepsilon} + \frac{17}{288} g_1^2 a_t^2 \right. \\
& - \frac{19}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{4} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + \frac{1}{4} \lambda_h a_t^2 + \frac{179}{576} g_2^2 a_t^2 + \frac{19}{48} \frac{g_2^2 a_t^2}{\varepsilon} \\
& \left. + \frac{1}{2} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} H_i D_{\alpha_2}^{\dot{\alpha}_2} H_j (H^\dagger)_k (H^\dagger)_l
\end{aligned} \tag{28}$$

term23

$$\begin{aligned}
& + \left(\frac{1}{96} dg_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_1^2 a_t^2 - \frac{1}{48} \kappa^2 + \frac{1}{48} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_1^2 a_t^2}{\varepsilon} - \frac{11}{288} dg_2^2 a_t^2 - \frac{1}{24} g_1^2 a_t^2 \right. \\
& - \frac{1}{24} \frac{dg_2^2 a_t^2}{\varepsilon} - \frac{5}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{5}{48} \frac{g_2^2 a_t^2}{\varepsilon} + \frac{1}{8} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{8} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{8} g_1^2 a_t^2 \hat{\mu}_h^2 \\
& + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{8} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{6} a_t^3 \hat{\mu} - \frac{3}{16} \varepsilon \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{121}{576} g_2^2 a_t^2 + \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}^2 \\
& - \frac{1}{4} \ln \frac{M_s^2}{\mu_M^2} a_t^4 + \frac{1}{4} \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2}{\varepsilon} - \frac{1}{4} \frac{g_1^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} - \frac{13}{48} a_t^2 \hat{\mu}^2 + \frac{11}{32} \varepsilon a_t^4 + \frac{5}{12} \kappa a_t^2 \\
& - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t \hat{\mu} - \frac{1}{2} \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - \frac{1}{2} \frac{a_t^2 \hat{\mu}^2}{\varepsilon} + \frac{1}{2} \frac{a_t^4}{\varepsilon} - \frac{1}{2} \frac{\lambda_h a_t^2}{\varepsilon} + \frac{13}{24} \kappa a_t \hat{\mu} - \frac{7}{8} \lambda_h a_t^2 + \frac{7}{6} a_t^4 \\
& \left. + \frac{5}{4} \kappa a_t^2 \hat{\mu}_h^2 + \frac{\kappa a_t \hat{\mu}}{\varepsilon} + \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \varepsilon_{ik} \varepsilon_{jl} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_l \quad (29)
\end{aligned}$$

term24

$$\begin{aligned}
& - \left(\frac{1}{96} dg_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{11}{576} dg_2^2 a_t^2 - \frac{1}{48} g_1^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{dg_2^2 a_t^2}{\varepsilon} + \frac{1}{24} \frac{g_1^2 a_t^2}{\varepsilon} + \frac{17}{288} g_1^2 a_t^2 \right. \\
& - \frac{19}{96} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{4} g_2^2 a_t^2 \hat{\mu}_h^2 - \frac{1}{4} g_2^2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + \frac{1}{4} \lambda_h a_t^2 + \frac{179}{576} g_2^2 a_t^2 + \frac{19}{48} \frac{g_2^2 a_t^2}{\varepsilon} \\
& \left. + \frac{1}{2} \frac{g_2^2 a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2} \varepsilon_{il} \varepsilon_{jk} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} H_i H_j D_{\alpha_1}^{\dot{\alpha}_1} (H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_2} (H^\dagger)_l \quad (30)
\end{aligned}$$

nD: 0, H: 2, H+: 2

term26

$$\begin{aligned}
& + \left(a_t^3 \hat{\mu} - \ln \frac{M_s^2}{\mu_M^2} a_t^3 \hat{\mu} \hat{\mu}_h^2 - \frac{1}{4} a_t^2 \hat{\mu}^2 - \frac{1}{4} \kappa^2 \ln \frac{M_s^2}{\mu_M^2} + \frac{1}{4} \lambda_\varphi a_t^2 - \frac{1}{4} \lambda_\varphi \ln \frac{M_s^2}{\mu_M^2} a_t^2 + \frac{1}{2} \kappa a_t \hat{\mu} + \frac{1}{2} \frac{\kappa^2}{\varepsilon} \right. \\
& + \frac{1}{2} \frac{\lambda_\varphi a_t^2}{\varepsilon} + 2 a_t^3 \hat{\mu} \hat{\mu}_h^2 + 2 \ln \frac{M_s^2}{\mu_M^2} a_t^4 - 2 \kappa a_t^2 \hat{\mu}_h^4 + 2 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 + 2 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^4 + 2 \frac{a_t^3 \hat{\mu} \hat{\mu}_h^2}{\varepsilon} - 3 a_t^4 \\
& - 3 \kappa a_t^2 + 3 \kappa \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 + 3 \lambda_h a_t^2 - 3 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 - 4 \kappa a_t^2 \hat{\mu}_h^2 - 4 \frac{a_t^4}{\varepsilon} - 4 \frac{\kappa a_t^2}{\varepsilon} - 4 \frac{\kappa a_t^2 \hat{\mu}_h^4}{\varepsilon} \\
& + 6 \ln \frac{M_s^2}{\mu_M^2} a_t^4 \hat{\mu}_h^2 + 6 \lambda_h a_t^2 \hat{\mu}_h^2 - 6 \lambda_h \ln \frac{M_s^2}{\mu_M^2} a_t^2 \hat{\mu}_h^2 - 6 \frac{\kappa a_t^2 \hat{\mu}_h^2}{\varepsilon} + 6 \frac{\lambda_h a_t^2}{\varepsilon} - 8 a_t^4 \hat{\mu}_h^2 - 12 \frac{a_t^4 \hat{\mu}_h^2}{\varepsilon} \\
& \left. + 12 \frac{\lambda_h a_t^2 \hat{\mu}_h^2}{\varepsilon} \right) \varepsilon_{ik} \varepsilon_{jl} H_i H_j (H^\dagger)_k (H^\dagger)_l \quad (31)
\end{aligned}$$

nD: 2, WL: 1, H: 1, H+: 1

term22

$$+ \left(\frac{1}{96} g_2 \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{48} \frac{g_2 a_t^2}{\varepsilon} - \frac{17}{576} g_2 a_t^2 \right) M_s^{-2} \varepsilon_{kj} \varepsilon_{il} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} \varepsilon^{\alpha_1 \alpha_4} \varepsilon^{\alpha_2 \alpha_3} (W_L)_{\alpha_1 \alpha_2, ij} D_{\alpha_3}^{\dot{\alpha}_1} H_k D_{\alpha_4}^{\dot{\alpha}_2} (H^\dagger)_l \quad (32)$$

term22

$$+\left(\frac{1}{96}g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_2a_t^2}{\varepsilon}-\frac{17}{576}g_2a_t^2\right)M_s^{-2}\varepsilon_{kj}\varepsilon_{il}\varepsilon_{\dot{\alpha}2\dot{\alpha}1}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(W_L)_{\alpha_1\alpha_2,ij}D_{\alpha_3}^{\dot{\alpha}1}H_kD_{\alpha_4}^{\dot{\alpha}2}(H^\dagger)_l \quad (33)$$

term25

$$+\left(\frac{1}{96}g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_2a_t^2}{\varepsilon}-\frac{17}{576}g_2a_t^2\right)M_s^{-2}\varepsilon_{kj}\varepsilon_{il}\varepsilon_{\dot{\alpha}2\dot{\alpha}1}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(W_L)_{\alpha_1\alpha_2,ij}D_{\alpha_3}^{\dot{\alpha}1}H_kD_{\alpha_4}^{\dot{\alpha}2}(H^\dagger)_l \quad (34)$$

term25

$$+\left(\frac{1}{96}g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_2a_t^2}{\varepsilon}-\frac{17}{576}g_2a_t^2\right)M_s^{-2}\varepsilon_{kj}\varepsilon_{il}\varepsilon_{\dot{\alpha}2\dot{\alpha}1}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(W_L)_{\alpha_1\alpha_2,ij}D_{\alpha_3}^{\dot{\alpha}1}H_kD_{\alpha_4}^{\dot{\alpha}2}(H^\dagger)_l \quad (35)$$

nD: 2, H: 1, H+: 1, WL+: 1

term22

$$-\left(\frac{1}{96}g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_2a_t^2}{\varepsilon}-\frac{17}{576}g_2a_t^2\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}3\dot{\alpha}2}\varepsilon_{\dot{\alpha}4\dot{\alpha}1}D_{\alpha_1}^{\dot{\alpha}1}H_iD_{\alpha_2}^{\dot{\alpha}2}(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}3\dot{\alpha}4} \quad (36)$$

term22

$$-\left(\frac{1}{96}g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_2a_t^2}{\varepsilon}-\frac{17}{576}g_2a_t^2\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}3\dot{\alpha}1}\varepsilon_{\dot{\alpha}4\dot{\alpha}2}D_{\alpha_1}^{\dot{\alpha}1}H_iD_{\alpha_2}^{\dot{\alpha}2}(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}3\dot{\alpha}4} \quad (37)$$

term25

$$-\left(\frac{1}{96}g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_2a_t^2}{\varepsilon}-\frac{17}{576}g_2a_t^2\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}3\dot{\alpha}2}\varepsilon_{\dot{\alpha}4\dot{\alpha}1}D_{\alpha_1}^{\dot{\alpha}1}H_iD_{\alpha_2}^{\dot{\alpha}2}(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}3\dot{\alpha}4} \quad (38)$$

term25

$$-\left(\frac{1}{96}g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{1}{48}\frac{g_2a_t^2}{\varepsilon}-\frac{17}{576}g_2a_t^2\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}3\dot{\alpha}1}\varepsilon_{\dot{\alpha}4\dot{\alpha}2}D_{\alpha_1}^{\dot{\alpha}1}H_iD_{\alpha_2}^{\dot{\alpha}2}(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}3\dot{\alpha}4} \quad (39)$$

nD: 0, H: 1, H+: 1, BL+: 2

term27

$$+\left(\frac{5}{576}g_1^2a_t^2-\frac{1}{96}g_1^2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{48}\frac{g_1^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon_{\dot{\alpha}1\dot{\alpha}3}\varepsilon_{\dot{\alpha}2\dot{\alpha}4}H_i(H^\dagger)_j(B_R)^{\dot{\alpha}1\dot{\alpha}2}(B_R)^{\dot{\alpha}3\dot{\alpha}4} \quad (40)$$

term27

$$+\left(\frac{5}{576}g_1^2a_t^2-\frac{1}{96}g_1^2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{48}\frac{g_1^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon_{\dot{\alpha}1\dot{\alpha}4}\varepsilon_{\dot{\alpha}2\dot{\alpha}3}H_i(H^\dagger)_j(B_R)^{\dot{\alpha}1\dot{\alpha}2}(B_R)^{\dot{\alpha}3\dot{\alpha}4} \quad (41)$$

nD: 0, BL: 2, H: 1, H+: 1

term27

$$+\left(\frac{5}{576}g_1^2a_t^2-\frac{1}{96}g_1^2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{48}\frac{g_1^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(B_L)_{\alpha_1\alpha_2}(B_L)_{\alpha_3\alpha_4}H_i(H^\dagger)_j \quad (42)$$

term27

$$+\left(\frac{5}{576}g_1^2a_t^2-\frac{1}{96}g_1^2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{48}\frac{g_1^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(B_L)_{\alpha_1\alpha_2}(B_L)_{\alpha_3\alpha_4}H_i(H^\dagger)_j \quad (43)$$

nD: 0, H: 1, H+: 1, BL+: 1, WL+: 1

term28

$$-\left(\frac{5}{288}g_1g_2a_t^2-\frac{1}{48}g_1g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{24}\frac{g_1g_2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_4}H_i(H^\dagger)_j(B_R)^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{kl}^{\dot{\alpha}_3\dot{\alpha}_4} \quad (44)$$

term28

$$-\left(\frac{5}{288}g_1g_2a_t^2-\frac{1}{48}g_1g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{24}\frac{g_1g_2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{il}\varepsilon_{kj}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_4}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_3}H_i(H^\dagger)_j(B_R)^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{kl}^{\dot{\alpha}_3\dot{\alpha}_4} \quad (45)$$

nD: 0, BL: 1, WL: 1, H: 1, H+: 1

term28

$$-\left(\frac{5}{288}g_1g_2a_t^2-\frac{1}{48}g_1g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{24}\frac{g_1g_2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{kj}\varepsilon_{il}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(B_L)_{\alpha_1\alpha_2}(W_L)_{\alpha_3\alpha_4,ij}H_k(H^\dagger)_l \quad (46)$$

term28

$$-\left(\frac{5}{288}g_1g_2a_t^2-\frac{1}{48}g_1g_2\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{24}\frac{g_1g_2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{kj}\varepsilon_{il}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(B_L)_{\alpha_1\alpha_2}(W_L)_{\alpha_3\alpha_4,ij}H_k(H^\dagger)_l \quad (47)$$

nD: 0, H: 3, H+: 3

term29

$$\begin{aligned} &-\left(a_t^4\hat{\mu}^2-\kappa a_t^3\hat{\mu}\hat{\mu}_h^4+\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu}+\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu}\hat{\mu}_h^4-\lambda_\varphi a_t^4\hat{\mu}_h^2-\frac{1}{12}\kappa^3-\frac{1}{12}\lambda_\varphi\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu}-\frac{1}{6}a_t^3\hat{\mu}^3\right. \\ &+\frac{1}{6}\frac{\lambda_\varphi a_t^3\hat{\mu}}{\varepsilon}-\frac{1}{4}\kappa^2a_t\hat{\mu}+\frac{1}{4}\lambda_\varphi\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^2+\frac{1}{3}\lambda_\varphi a_t^3\hat{\mu}+\frac{1}{2}\kappa a_t^2\hat{\mu}^2-\frac{1}{2}\lambda_\varphi a_t^4+\frac{1}{2}\lambda_\varphi\ln\frac{M_s^2}{\mu_M^2}a_t^4\hat{\mu}_h^2 \\ &-\frac{1}{2}\lambda_\varphi\kappa a_t^2-\frac{1}{2}\frac{\lambda_\varphi\kappa a_t^2}{\varepsilon}-\frac{3}{2}\ln\frac{M_s^2}{\mu_M^2}a_t^4\hat{\mu}^2\hat{\mu}_h^2+\frac{7}{2}a_t^4\hat{\mu}^2\hat{\mu}_h^2+\frac{7}{2}\ln\frac{M_s^2}{\mu_M^2}a_t^5\hat{\mu}+\frac{11}{2}\kappa^2a_t^2-\frac{15}{2}a_t^5\hat{\mu} \\ &+\frac{19}{2}\kappa^2a_t^2\hat{\mu}_h^2-\frac{21}{2}\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^4+\frac{43}{3}a_t^6+\frac{37}{2}\kappa a_t^4-2\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu}-2\frac{\kappa a_t^3\hat{\mu}}{\varepsilon}-2\frac{\kappa a_t^3\hat{\mu}\hat{\mu}_h^4}{\varepsilon} \end{aligned}$$

$$\begin{aligned}
& + 3a_t^6\hat{\mu}_h^2 - 3\ln\frac{M_s^2}{\mu_M^2}a_t^6\hat{\mu}_h^2 + 3\kappa^2a_t^2\hat{\mu}_h^4 - 3\kappa^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 - 3\kappa^2\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^4 + 3\frac{a_t^4\hat{\mu}^2\hat{\mu}_h^2}{\varepsilon} + 4\lambda_h a_t^3\hat{\mu} \\
& + 4\lambda_h^2 a_t^2 - 4\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + 4\frac{\lambda_h a_t^3\hat{\mu}}{\varepsilon} + 5\ln\frac{M_s^2}{\mu_M^2}a_t^5\hat{\mu}\hat{\mu}_h^2 - 5\kappa a_t^3\hat{\mu} + 6\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu}\hat{\mu}_h^2 \\
& - 6\kappa^2\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 + 6\frac{a_t^6\hat{\mu}_h^2}{\varepsilon} + 6\frac{\kappa^2 a_t^2}{\varepsilon} + 6\frac{\kappa^2 a_t^2\hat{\mu}_h^4}{\varepsilon} - 7a_t^5\hat{\mu}\hat{\mu}_h^2 - 7\frac{a_t^5\hat{\mu}}{\varepsilon} - 8\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu}\hat{\mu}_h^2 \\
& + 8\lambda_h\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^2 + 8\frac{\lambda_h^2 a_t^2}{\varepsilon} - 9\ln\frac{M_s^2}{\mu_M^2}a_t^6 - 10\lambda_h\kappa a_t^2 - 10\frac{a_t^5\hat{\mu}\hat{\mu}_h^2}{\varepsilon} - 12\kappa a_t^3\hat{\mu}\hat{\mu}_h^2 + 12\lambda_h a_t^3\hat{\mu}\hat{\mu}_h^2 \\
& - 12\lambda_h a_t^4\hat{\mu}_h^2 + 12\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^4 + 12\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^4\hat{\mu}_h^2 + 12\lambda_h^2 a_t^2\hat{\mu}_h^2 - 12\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 \\
& - 12\frac{\kappa a_t^3\hat{\mu}\hat{\mu}_h^2}{\varepsilon} + 12\frac{\kappa^2 a_t^2\hat{\mu}_h^2}{\varepsilon} - 16\lambda_h a_t^4 + 16\frac{\lambda_h a_t^3\hat{\mu}\hat{\mu}_h^2}{\varepsilon} - 16\frac{\lambda_h\kappa a_t^2}{\varepsilon} + 18\frac{a_t^6}{\varepsilon} + 20\lambda_h\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 \\
& + 21\frac{\kappa a_t^4}{\varepsilon} - 24\lambda_h\kappa a_t^2\hat{\mu}_h^2 - 24\frac{\lambda_h a_t^4}{\varepsilon} - 24\frac{\lambda_h a_t^4\hat{\mu}_h^2}{\varepsilon} + 24\frac{\lambda_h^2 a_t^2\hat{\mu}_h^2}{\varepsilon} - 27\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^4\hat{\mu}_h^2 + 37\kappa a_t^4\hat{\mu}_h^2 \\
& - 40\frac{\lambda_h\kappa a_t^2\hat{\mu}_h^2}{\varepsilon} + 54\frac{\kappa a_t^4\hat{\mu}_h^2}{\varepsilon} - \frac{\lambda_\varphi a_t^4\hat{\mu}_h^2}{\varepsilon} \Big) M_s^{-2}\varepsilon_{il}\varepsilon_{jm}\varepsilon_{kn}H_iH_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (48)
\end{aligned}$$

term30

$$\begin{aligned}
& + \left(\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu} - \lambda_h\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^2 - 2\lambda_h a_t^3\hat{\mu} + 2\lambda_h\kappa a_t^2 - 2\lambda_h^2 a_t^2 + 2\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 - 2\frac{\lambda_h a_t^3\hat{\mu}}{\varepsilon} \right. \\
& + 2\frac{\lambda_h\kappa a_t^2}{\varepsilon} + 4\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^3\hat{\mu}\hat{\mu}_h^2 - 4\lambda_h\kappa\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 - 4\frac{\lambda_h^2 a_t^2}{\varepsilon} - 6\lambda_h a_t^3\hat{\mu}\hat{\mu}_h^2 + 6\lambda_h a_t^4\hat{\mu}_h^2 \\
& - 6\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^4 - 6\lambda_h\ln\frac{M_s^2}{\mu_M^2}a_t^4\hat{\mu}_h^2 + 6\lambda_h\kappa a_t^2\hat{\mu}_h^2 - 6\lambda_h^2 a_t^2\hat{\mu}_h^2 + 6\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 + 8\lambda_h a_t^4 \\
& \left. - 8\frac{\lambda_h a_t^3\hat{\mu}\hat{\mu}_h^2}{\varepsilon} + 8\frac{\lambda_h\kappa a_t^2\hat{\mu}_h^2}{\varepsilon} + 12\frac{\lambda_h a_t^4}{\varepsilon} + 12\frac{\lambda_h a_t^4\hat{\mu}_h^2}{\varepsilon} - 12\frac{\lambda_h^2 a_t^2\hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2}\varepsilon_{im}\varepsilon_{jl}\varepsilon_{kn}H_iH_jH_k(H^\dagger)_l \\
& (H^\dagger)_m(H^\dagger)_n \quad (49)
\end{aligned}$$

term31

$$\begin{aligned}
& - \left(2\lambda_h^2 a_t^2 - 2\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + 4\frac{\lambda_h^2 a_t^2}{\varepsilon} + 6\lambda_h^2 a_t^2\hat{\mu}_h^2 - 6\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 + 12\frac{\lambda_h^2 a_t^2\hat{\mu}_h^2}{\varepsilon} \right) M_s^{-2}\varepsilon_{il}\varepsilon_{jn}\varepsilon_{km}H_i \\
& H_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (50)
\end{aligned}$$

term32

$$\begin{aligned}
& - \left(\frac{1}{2}\lambda_h^2 a_t^2 - \frac{1}{2}\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{3}{2}\lambda_h^2 a_t^2\hat{\mu}_h^2 - \frac{3}{2}\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 + 3\frac{\lambda_h^2 a_t^2\hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 a_t^2}{\varepsilon} \right) M_s^{-2}\varepsilon_{in}\varepsilon_{jl}\varepsilon_{km}H_i \\
& H_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (51)
\end{aligned}$$

term33

$$\begin{aligned}
& - \left(\frac{1}{2}\lambda_h^2 a_t^2 - \frac{1}{2}\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{3}{2}\lambda_h^2 a_t^2\hat{\mu}_h^2 - \frac{3}{2}\lambda_h^2\ln\frac{M_s^2}{\mu_M^2}a_t^2\hat{\mu}_h^2 + 3\frac{\lambda_h^2 a_t^2\hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 a_t^2}{\varepsilon} \right) M_s^{-2}\varepsilon_{im}\varepsilon_{jn}\varepsilon_{kl}H_i \\
& H_jH_k(H^\dagger)_l(H^\dagger)_m(H^\dagger)_n \quad (52)
\end{aligned}$$

nD: 0, H: 1, H+: 1, WL+: 2

term34

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{il}\varepsilon_{mj}\varepsilon_{nk}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_4}H_i(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{mn}^{\dot{\alpha}_3\dot{\alpha}_4} \quad (53)$$

term34

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{il}\varepsilon_{mj}\varepsilon_{nk}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_4}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_3}H_i(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{mn}^{\dot{\alpha}_3\dot{\alpha}_4} \quad (54)$$

term35

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{in}\varepsilon_{kj}\varepsilon_{lm}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_3}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_4}H_i(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{mn}^{\dot{\alpha}_3\dot{\alpha}_4} \quad (55)$$

term35

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{in}\varepsilon_{kj}\varepsilon_{lm}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_4}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_3}H_i(H^\dagger)_j(W_R)_{kl}^{\dot{\alpha}_1\dot{\alpha}_2}(W_R)_{mn}^{\dot{\alpha}_3\dot{\alpha}_4} \quad (56)$$

nD: 0, WL: 2, H: 1, H+: 1

term34

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{mj}\varepsilon_{kn}\varepsilon_{li}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(W_L)_{\alpha_1\alpha_2,ij}(W_L)_{\alpha_3\alpha_4,kl}H_m(H^\dagger)_n \quad (57)$$

term34

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{mj}\varepsilon_{kn}\varepsilon_{li}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(W_L)_{\alpha_1\alpha_2,ij}(W_L)_{\alpha_3\alpha_4,kl}H_m(H^\dagger)_n \quad (58)$$

term35

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{ml}\varepsilon_{in}\varepsilon_{jk}\varepsilon^{\alpha_1\alpha_3}\varepsilon^{\alpha_2\alpha_4}(W_L)_{\alpha_1\alpha_2,ij}(W_L)_{\alpha_3\alpha_4,kl}H_m(H^\dagger)_n \quad (59)$$

term35

$$-\left(\frac{5}{1152}g_2^2a_t^2 - \frac{1}{192}g_2^2\ln\frac{M_s^2}{\mu_M^2}a_t^2 + \frac{1}{96}\frac{g_2^2a_t^2}{\varepsilon}\right)M_s^{-2}\varepsilon_{ml}\varepsilon_{in}\varepsilon_{jk}\varepsilon^{\alpha_1\alpha_4}\varepsilon^{\alpha_2\alpha_3}(W_L)_{\alpha_1\alpha_2,ij}(W_L)_{\alpha_3\alpha_4,kl}H_m(H^\dagger)_n \quad (60)$$

nD: 1, L: 1, H: 1, H+: 1, L+: 1

term38

$$-\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon} - \frac{5}{8}a_t^2\right)M_s^{-2}(y_e)^{wu}(y_e^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}L_{\alpha_1,i}^vD_{\alpha_2}^{\dot{\alpha}_1}H_j(H^\dagger)_k(L^\dagger)_l^{\dot{\alpha}_2,w} \quad (61)$$

term43

$$+\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon} - \frac{5}{8}a_t^2\right)M_s^{-2}(y_e)^{wu}(y_e^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}L_{\alpha_1,i}^vH_jD_{\alpha_2}^{\dot{\alpha}_1}(H^\dagger)_k(L^\dagger)_l^{\dot{\alpha}_2,w} \quad (62)$$

term48

$$-\left(\frac{1}{4}a_t^2\right)M_s^{-2}(y_e)^{wu}(y_e^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}L_{\alpha_2,i}^vH_j(H^\dagger)_k(L^\dagger)_l^{\dot{\alpha}_2,w} \quad (63)$$

term53

$$-\left(\frac{1}{4}a_t^2\right)M_s^{-2}(y_e)^{wu}(y_e^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}L_{\alpha_1,i}^vH_j(H^\dagger)_kD_{\alpha_2}^{\dot{\alpha}_1}(L^\dagger)_l^{\dot{\alpha}_2,w} \quad (64)$$

term58

$$-\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{8}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{wu}(y_e^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}L_{\alpha_1,i}^vH_j(H^\dagger)_kD_{\alpha_2}^{\dot{\alpha}_1}(L^\dagger)_l^{\dot{\alpha}_2,w} \quad (65)$$

term63

$$-\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{8}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{wu}(y_e^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}L_{\alpha_2,i}^vH_j(H^\dagger)_k(L^\dagger)_l^{\dot{\alpha}_2,w} \quad (66)$$

nD: 1, Q: 1, H: 1, H+: 1, Q+: 1

term39

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon_{abc}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}Q_{\alpha_1,i,a}^vD_{\alpha_2}^{\dot{\alpha}_1}H_j(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (67)$$

term44

$$+\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon_{abc}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}Q_{\alpha_1,i,a}^vH_jD_{\alpha_2}^{\dot{\alpha}_1}(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (68)$$

term49

$$+\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}Q_{\alpha_2,i,a}^vH_j(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (69)$$

term54

$$-\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon_{abc}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}Q_{\alpha_1,i,a}^vH_j(H^\dagger)_kD_{\alpha_2}^{\dot{\alpha}_1}(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (70)$$

term59

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}Q_{\alpha_1,i,a}^v H_j(H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1}(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (71)$$

term64

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{jl}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}Q_{\alpha_2,i,a}^v H_j(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (72)$$

term88

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}\varepsilon_{ji}\varepsilon_{kl}\varepsilon_{abc}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}Q_{\alpha_1,i,a}^v D_{\alpha_2}^{\dot{\alpha}_1}H_j(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (73)$$

term89

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}\varepsilon_{ij}\varepsilon_{kl}\varepsilon_{abc}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}Q_{\alpha_1,i,a}^v H_j D_{\alpha_2}^{\dot{\alpha}_1}(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (74)$$

term90

$$+\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}\varepsilon_{ji}\varepsilon_{lk}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}Q_{\alpha_2,i,a}^v H_j(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (75)$$

term91

$$+\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}\varepsilon_{ij}\varepsilon_{lk}\varepsilon_{abc}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}Q_{\alpha_1,i,a}^v H_j(H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1}(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (76)$$

term92

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}\varepsilon_{ij}\varepsilon_{kl}\varepsilon_{abc}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}Q_{\alpha_1,i,a}^v H_j(H^\dagger)_k D_{\alpha_2}^{\dot{\alpha}_1}(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (77)$$

term93

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}\varepsilon_{ji}\varepsilon_{lk}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}Q_{\alpha_2,i,a}^v H_j(H^\dagger)_k(Q^\dagger)_{l,bc}^{\dot{\alpha}_2,w} \quad (78)$$

nD: 1, eC: 1, H: 1, H+: 1, eC+: 1

term40

$$+\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon} - \frac{5}{8}a_t^2\right)M_s^{-2}(y_e)^{uw}(y_e^\dagger)^{vu}\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(e_C)_\alpha^v D_{\alpha_2}^{\dot{\alpha}_1}H_i(H^\dagger)_j(e_C^\dagger)^{\dot{\alpha}_2,w} \quad (79)$$

term46

$$-\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon} - \frac{5}{8}a_t^2\right)M_s^{-2}(y_e)^{uw}(y_e^\dagger)^{vu}\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(e_C)_\alpha^v H_i D_{\alpha_2}^{\dot{\alpha}_1}(H^\dagger)_j(e_C^\dagger)^{\dot{\alpha}_2,w} \quad (80)$$

term51

$$-\left(\frac{1}{4}a_t^2\right)M_s^{-2}(y_e)^{uw}(y_e^\dagger)^{vu}\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(e_{\mathbb{C}})^v_{\alpha_1}H_i(H^\dagger)_jD_{\alpha_2}^{\dot{\alpha}_1}(e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2,w} \quad (81)$$

term55

$$-\left(\frac{1}{4}a_t^2\right)M_s^{-2}(y_e)^{uw}(y_e^\dagger)^{vu}\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}(e_{\mathbb{C}})^v_{\alpha_2}H_i(H^\dagger)_j(e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2,w} \quad (82)$$

term61

$$-\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{8}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{uw}(y_e^\dagger)^{vu}\varepsilon_{ij}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}(e_{\mathbb{C}})^v_{\alpha_2}H_i(H^\dagger)_j(e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2,w} \quad (83)$$

term65

$$-\left(\frac{1}{4}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{8}a_t^2 - \frac{1}{2}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{uw}(y_e^\dagger)^{vu}\varepsilon_{ij}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(e_{\mathbb{C}})^v_{\alpha_1}H_i(H^\dagger)_jD_{\alpha_2}^{\dot{\alpha}_1}(e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2,w} \quad (84)$$

nD: 1, dC: 1, H: 1, H+: 1, dC+: 1

term41

$$+\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_d)^{uw}(y_d^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(d_{\mathbb{C}})^v_{\alpha_1,ab}D_{\alpha_2}^{\dot{\alpha}_1}H_i(H^\dagger)_j(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (85)$$

term47

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_d)^{uw}(y_d^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(d_{\mathbb{C}})^v_{\alpha_1,ab}H_iD_{\alpha_2}^{\dot{\alpha}_1}(H^\dagger)_j(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (86)$$

term52

$$+\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_d)^{uw}(y_d^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(d_{\mathbb{C}})^v_{\alpha_1,ab}H_i(H^\dagger)_jD_{\alpha_2}^{\dot{\alpha}_1}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (87)$$

term56

$$-\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_d)^{uw}(y_d^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}D_{\alpha_1}^{\dot{\alpha}_1}(d_{\mathbb{C}})^v_{\alpha_2,ab}H_i(H^\dagger)_j(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (88)$$

term62

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{uw}(y_d^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}(d_{\mathbb{C}})^v_{\alpha_2,ab}H_i(H^\dagger)_j(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (89)$$

term66

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{uw}(y_d^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(d_{\mathbb{C}})^v_{\alpha_1,ab}H_i(H^\dagger)_jD_{\alpha_2}^{\dot{\alpha}_1}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (90)$$

nD: 1, uC: 1, H: 1, H+: 1, uC+: 1

term42

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_u)^{uw}(y_u^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(u_{\mathbb{C}})^v_{\alpha_1,ab}D_{\alpha_2}^{\dot{\alpha}_1}H_i(H^\dagger)_j(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (91)$$

term45

$$+\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_u)^{uw}(y_u^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(u_{\mathbb{C}})^v_{\alpha_1,ab}H_iD_{\alpha_2}^{\dot{\alpha}_1}(H^\dagger)_j(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (92)$$

term50

$$+\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_u)^{uw}(y_u^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(u_{\mathbb{C}})^v_{\alpha_1,ab}H_i(H^\dagger)_jD_{\alpha_2}^{\dot{\alpha}_1}(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (93)$$

term57

$$-\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_u)^{uw}(y_u^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}(u_{\mathbb{C}})^v_{\alpha_2,ab}H_i(H^\dagger)_j(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (94)$$

term60

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{uw}(y_u^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}D_{\alpha_1}^{\dot{\alpha}_1}(u_{\mathbb{C}})^v_{\alpha_2,ab}H_i(H^\dagger)_j(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (95)$$

term67

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{16}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{uw}(y_u^\dagger)^{vu}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(u_{\mathbb{C}})^v_{\alpha_1,ab}H_i(H^\dagger)_jD_{\alpha_2}^{\dot{\alpha}_1}(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (96)$$

nD: 1, uC: 1, H: 2, dC+: 1

term68

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_u^\dagger)^{vu}(y_d)^{uw}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(u_{\mathbb{C}})^v_{\alpha_1,ab}D_{\alpha_2}^{\dot{\alpha}_1}H_iH_j(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (97)$$

term69

$$-\left(\frac{1}{8}\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{1}{4}\frac{a_t^2}{\varepsilon} - \frac{5}{16}a_t^2\right)M_s^{-2}(y_u^\dagger)^{vu}(y_d)^{uw}\varepsilon_{ij}\varepsilon_{cab}\varepsilon^{\alpha_2\alpha_1}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(u_{\mathbb{C}})^v_{\alpha_1,ab}D_{\alpha_2}^{\dot{\alpha}_1}H_iH_j(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (98)$$

term73

$$+\left(\frac{1}{8}a_t^2\right)M_s^{-2}(y_u^\dagger)^{vu}(y_d)^{uw}\varepsilon_{ji}\varepsilon_{cab}\varepsilon^{\alpha_1\alpha_2}\varepsilon_{\dot{\alpha}_1\dot{\alpha}_2}(u_{\mathbb{C}})^v_{\alpha_1,ab}H_iH_jD_{\alpha_2}^{\dot{\alpha}_1}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,w} \quad (99)$$

term74

$$+ \left(\frac{1}{8} a_t^2 \right) M_s^{-2} (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_2 \dot{\alpha}_1} D_{\alpha_1}^{\dot{\alpha}_1} (u_{\mathbb{C}})^v_{\alpha_2, ab} H_i H_j (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (100)$$

term77

$$- \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (u_{\mathbb{C}})^v_{\alpha_2, ab} H_i H_j (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (101)$$

term78

$$+ \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} (y_u^\dagger)^{vu} (y_d)^{uw} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (u_{\mathbb{C}})^v_{\alpha_1, ab} H_i H_j D_{\alpha_2}^{\dot{\alpha}_1} (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (102)$$

nD: 1, dC: 1, H+: 2, uC+: 1

term70

$$+ \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (103)$$

term71

$$- \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} - \frac{5}{16} a_t^2 \right) M_s^{-2} (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} D_{\alpha_2}^{\dot{\alpha}_1} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (104)$$

term72

$$- \left(\frac{1}{8} a_t^2 \right) M_s^{-2} (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_1 \alpha_2} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} (H^\dagger)_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (105)$$

term75

$$- \left(\frac{1}{8} a_t^2 \right) M_s^{-2} (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ij} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (d_{\mathbb{C}})^v_{\alpha_2, ab} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (106)$$

term76

$$- \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} D_{\alpha_1}^{\dot{\alpha}_1} (d_{\mathbb{C}})^v_{\alpha_2, ab} (H^\dagger)_i (H^\dagger)_j (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (107)$$

term79

$$+ \left(\frac{1}{8} \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{16} a_t^2 - \frac{1}{4} \frac{a_t^2}{\varepsilon} \right) M_s^{-2} (y_u)^{uw} (y_d^\dagger)^{vu} \varepsilon_{ji} \varepsilon_{cab} \varepsilon^{\alpha_2 \alpha_1} \varepsilon_{\dot{\alpha}_1 \dot{\alpha}_2} (d_{\mathbb{C}})^v_{\alpha_1, ab} (H^\dagger)_i (H^\dagger)_j D_{\alpha_2}^{\dot{\alpha}_1} (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w} \quad (108)$$

nD: 0, L: 1, eC: 1, H: 1, H+: 2

term80

$$+\left(\frac{1}{8}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{16}da_t^2-\frac{1}{4}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{uv}(y_e^\dagger)^{xu}(y_e^\dagger)^{vw}\varepsilon_{il}\varepsilon_{jk}\varepsilon^{\alpha_1\alpha_2}L_{\alpha_1,i}^w(e_{\mathbb{C}})^x_{\alpha_2}H_j(H^\dagger)_k(H^\dagger)_l \quad (109)$$

term86

$$+\left(\frac{1}{8}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{16}da_t^2-\frac{1}{4}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{uv}(y_e^\dagger)^{xu}(y_e^\dagger)^{vw}\varepsilon_{jk}\varepsilon_{il}\varepsilon^{\alpha_1\alpha_2}L_{\alpha_1,i}^w(e_{\mathbb{C}})^x_{\alpha_2}H_j(H^\dagger)_k(H^\dagger)_l \quad (110)$$

nD: 0, Q: 1, dC: 1, H: 1, H+: 2

term81

$$+\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{32}da_t^2-\frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{uv}(y_d^\dagger)^{xu}(y_d^\dagger)^{vw}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}Q_{\alpha_1,i,a}^w(d_{\mathbb{C}})^x_{\alpha_2,bc}H_j(H^\dagger)_k(H^\dagger)_l \quad (111)$$

term87

$$+\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{32}da_t^2-\frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{uv}(y_d^\dagger)^{xu}(y_d^\dagger)^{vw}\varepsilon_{jk}\varepsilon_{il}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}Q_{\alpha_1,i,a}^w(d_{\mathbb{C}})^x_{\alpha_2,bc}H_j(H^\dagger)_k(H^\dagger)_l \quad (112)$$

term102

$$+\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{32}da_t^2-\frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{vu}(y_u^\dagger)^{uw}(y_d^\dagger)^{xv}\varepsilon_{ji}\varepsilon_{lk}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}Q_{\alpha_1,i,a}^w(d_{\mathbb{C}})^x_{\alpha_2,bc}H_j(H^\dagger)_k(H^\dagger)_l \quad (113)$$

term105

$$+\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{32}da_t^2-\frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{vu}(y_u^\dagger)^{uw}(y_d^\dagger)^{xv}\varepsilon_{ji}\varepsilon_{kl}\varepsilon_{abc}\varepsilon^{\alpha_1\alpha_2}Q_{\alpha_1,i,a}^w(d_{\mathbb{C}})^x_{\alpha_2,bc}H_j(H^\dagger)_k(H^\dagger)_l \quad (114)$$

nD: 0, H: 2, H+: 1, L+: 1, eC+: 1

term82

$$-\left(\frac{1}{8}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{16}da_t^2-\frac{1}{4}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{wu}(y_e)^{vx}(y_e^\dagger)^{uv}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(L^\dagger)_l^{\dot{\alpha}_1,w}(e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2,x} \quad (115)$$

term84

$$-\left(\frac{1}{8}d\ln\frac{M_s^2}{\mu_M^2}a_t^2-\frac{3}{16}da_t^2-\frac{1}{4}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_e)^{wu}(y_e)^{vx}(y_e^\dagger)^{uv}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(L^\dagger)_l^{\dot{\alpha}_1,w}(e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2,x} \quad (116)$$

nD: 0, H: 2, H+: 1, Q+: 1, dC+: 1

term83

$$-\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{wu}(y_d)^{vx}(y_d^\dagger)^{uv}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (117)$$

term85

$$-\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_d)^{wu}(y_d)^{vx}(y_d^\dagger)^{uv}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (118)$$

term103

$$+\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}(y_d)^{vx}\varepsilon_{ji}\varepsilon_{kl}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (119)$$

term104

$$+\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{wu}(y_u^\dagger)^{uv}(y_d)^{vx}\varepsilon_{ji}\varepsilon_{lk}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_iH_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (120)$$

nD: 0, H: 1, H+: 2, Q+: 1, uC+: 1

term94

$$+\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{wu}(y_u)^{vx}(y_u^\dagger)^{uv}\varepsilon_{ij}\varepsilon_{kl}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (121)$$

term95

$$-\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{vx}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{il}\varepsilon_{jk}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (122)$$

term99

$$-\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{wu}(y_u)^{vx}(y_u^\dagger)^{uv}\varepsilon_{ik}\varepsilon_{lj}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (123)$$

term101

$$-\left(\frac{1}{16}d\ln\frac{M_s^2}{\mu_M^2}a_t^2 - \frac{3}{32}da_t^2 - \frac{1}{8}\frac{da_t^2}{\varepsilon}\right)M_s^{-2}(y_u)^{vx}(y_d)^{wu}(y_d^\dagger)^{uv}\varepsilon_{il}\varepsilon_{kj}\varepsilon_{cab}\varepsilon_{\dot{\alpha}_2\dot{\alpha}_1}H_i(H^\dagger)_j(H^\dagger)_k(Q^\dagger)_{l,ab}^{\dot{\alpha}_1,w}(u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2,x} \quad (124)$$

nD: 0, Q: 1, uC: 1, H: 2, H+: 1

term96

$$+ \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} (y_u^\dagger)^{xu} (y_d)^{uv} (y_d^\dagger)^{vw} \varepsilon_{il} \varepsilon_{jk} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (u_C)^x_{\alpha_2, bc} H_j H_k (H^\dagger)_l \quad (125)$$

term97

$$+ \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} (y_u)^{uv} (y_u^\dagger)^{xu} (y_u^\dagger)^{vw} \varepsilon_{jl} \varepsilon_{ki} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (u_C)^x_{\alpha_2, bc} H_j H_k (H^\dagger)_l \quad (126)$$

term98

$$+ \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} (y_u)^{uv} (y_u^\dagger)^{xu} (y_u^\dagger)^{vw} \varepsilon_{ji} \varepsilon_{kl} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (u_C)^x_{\alpha_2, bc} H_j H_k (H^\dagger)_l \quad (127)$$

term100

$$+ \left(\frac{1}{16} d \ln \frac{M_s^2}{\mu_M^2} a_t^2 - \frac{3}{32} da_t^2 - \frac{1}{8} \frac{da_t^2}{\varepsilon} \right) M_s^{-2} (y_u^\dagger)^{xu} (y_d)^{uv} (y_d^\dagger)^{vw} \varepsilon_{il} \varepsilon_{kj} \varepsilon_{abc} \varepsilon^{\alpha_1 \alpha_2} Q_{\alpha_1, i, a}^w (u_C)^x_{\alpha_2, bc} H_j H_k (H^\dagger)_l \quad (128)$$